YESWANTH ARASAVALLI

Data Analyst | Data Scientist | Machine Learning Engineer
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SUMMARY

Computer Science postgraduate specializing in Big Data Analytics, with internship experience in SQL, Python, ETL pipelines, data cleaning, and Power BI/Tableau reporting, alongside freelance experience in AI data annotation and prompt evaluation. Reduced SQL query runtime by 25% and manual reporting effort by 40%. Applied data analytics, statistical modeling, and machine learning across projects in decision intelligence, predictive maintenance, NLP, computer vision, and order/revenue analytics, including local LLM-powered workflows.

SKILLS

Programming: SQL (CTEs, window functions, query optimization), Python, R

Analytics & BI: Power BI, Tableau, Excel, KPI analysis, dashboard development, exploratory data analysis (EDA), statistical analysis, hypothesis testing

Data Engineering: Pandas, NumPy, ETL pipelines, data cleaning and transformation, feature engineering, PostgreSQL, MongoDB, Apache Kafka, Apache Spark

Machine Learning: Scikit-learn, XGBoost, Random Forest, Logistic Regression, classification, segmentation, time-series forecasting, model evaluation

Deep Learning & NLP/CV: PyTorch, LSTM, DistilBERT, Hugging Face Transformers, TF-IDF, EfficientNet-B0, NLP, image classification

Tools & Cloud: FastAPI, Flask, REST APIs, Docker, MLflow, Evidently, Streamlit, Gradio, Ollama, Playwright, Git; AWS/GCP/Azure (project/certification exposure)

EXPERIENCE

AI Data Annotation Specialist & Prompt Engineer *Feb 2024 – Present*
TELUS International | Freelance / Project-Based | Remote

- Review NLP and computer-vision annotation batches, resolve ambiguous cases, and document labeling rules across review cycles.
- Evaluate prompt variants for generative-AI tasks and track output quality to identify recurring failure patterns.
- Flag systematic labeling errors and recommend structural fixes before training-data ingestion.

Data Science Intern *Jan 2024 – Feb 2024*
Postulate IRT, India

- Supported NLP speech-data preparation using Python, including data cleaning and consistency checks.
- Contributed to forecasting analysis and model evaluation to compare predictive performance against a baseline.
- Prepared Tableau dashboards to visualize KPIs and support recurring performance reporting.

Data Analyst Intern *Jul 2023 – Jan 2024*
Edunet Foundation, India

- Consolidated cloud and on-premises data with SQL pipelines using CTEs and window functions, reducing query runtime by 25%.
- Developed and evaluated tree-based customer-segmentation models using precision, recall, and F1-score to assess model performance.
- Built Power BI dashboards for operational KPI monitoring and collaborated with stakeholders to review metric definitions and reporting requirements.

Data Science Intern *Apr 2023 – Jul 2023*
PHN Technology Pvt. Ltd., India

- Automated Python/SQL ETL scripts for log-data ingestion, eliminating 3–4 hours of manual exports per week.
- Standardized Tableau and Excel reports, cutting manual reporting effort by 40%.
- Implemented and deployed R-based statistical models through IBM Watson Studio for repeatable internal analytics workflows.

PROJECTS

Domain-Agnostic Decision Intelligence System
Python, Scikit-learn, Pandas, Streamlit, Flask

- Architected a multi-stage analytics pipeline with data-quality checks, domain detection across 13 domains, KPI computation, driver-graph analysis, and scenario simulation to produce a 10-section report.
- Exposed pipeline functionality through a Flask REST API with 7 endpoints; validated with 454 test functions.

Order Intelligence Platform – Order & Revenue Analytics
Python, Apache Kafka, Apache Spark, PostgreSQL, Streamlit

- Created an interactive order/revenue analytics dashboard with region and currency filters, time-range analysis, and business KPI visualizations.
- Added reproducible synthetic-data demos, data-quality checks, and pipeline verification; the demo does not use live commercial data.

Predictive Maintenance – Machine Failure Forecasting
Python, Gradient Boosting, Time-Series ML, NASA C-MAPSS FD004

- Engineered a leakage-safe pipeline with causal temporal features and engine-grouped validation to estimate remaining useful life (RUL) and predict 30-cycle failure risk.
- Reported 31.86-cycle RUL MAE, 0.898 F1, and 0.974 PR-AUC with Gradient Boosting; added cost-sensitive threshold analysis and reproducibility checks.

Amazon Food Reviews Sentiment Analysis
Python, DistilBERT, PyTorch, Hugging Face, Gradio

- Fine-tuned DistilBERT on 568K+ reviews, reaching **~94%** accuracy versus ~89% for a TF-IDF + Logistic Regression baseline; evaluated with F1, ROC, and confusion matrix.
- Deployed an interactive Gradio demo on Hugging Face Spaces.

Retail Product Image Recognition System
Python, EfficientNet-B0, PyTorch, MLflow, Streamlit

- Classified ~200 SKU categories with a fine-tuned EfficientNet-B0, achieving **84.9%** Top-1 accuracy versus a 76% ResNet-18 baseline.
- Tracked experiments in MLflow and built a Streamlit interface for product prediction and confidence inspection.

AI Job Application Intelligence System
Python, Ollama, Streamlit, SQLite, FastAPI, Playwright

- Built a fully local workflow with no cloud dependencies for job-description processing, LLM-assisted resume tailoring, 6-dimension ATS scoring, and APPLY/MAYBE/IMPROVE recommendations.
- Produced an 8-page Streamlit dashboard and a Playwright-based job-listing scraper.

EDUCATION

M.S. Computer Science (Big Data Analytics) *2021 – 2023*
Central University of Rajasthan, Ajmer | GPA: 7.3/10
Capstone: Telecom churn prediction using an XGBoost + Random Forest ensemble, reaching 88% accuracy on 200K+ records.

B.S. Computer Science *2018 – 2021*
Adikavi Nannaya University, Rajahmundry | GPA: 8.37/10

CERTIFICATIONS

- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Google Advanced Data Analytics (Coursera)
- Data Science Specialization, Johns Hopkins University (Coursera)
- MongoDB Python Developer Associate (MongoDB University)
- Microsoft Power BI (Infosys Springboard)
- Machine Learning / AI Engineer (Skillsoft & Infosys Springboard)